ModelIt!

Teacher's Guide

Can We Capture Carbon from Thin Air?

Modeling Direct Air Capture Systems, Energy Requirements, and Scalability

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-ESS3-4, HS-PS1-4, HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-4, HS-PS1-4, HS-ETS1-3

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students evaluate claims about direct air capture as a climate solution using quantitative evidence from their models about energy requirements, costs, and scale challenges.
Disciplinary Core Idea	**ESS3.D Global Climate Change / PS1.B Chemical Reactions** — Students model how chemical sorbent reactions capture and release CO2 and evaluate direct air capture as a technological response to anthropogenic climate change.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students analyze the enormous scale mismatch between current DAC capacity (thousands of tonnes per year) and climate-relevant removal (billions of tonnes per year), using quantitative reasoning to evaluate feasibility.

CAST Testing Alignment

MULTIPLE CHOICE:

A direct air capture facility powered by natural gas with a carbon intensity of 0.4 tonnes CO2 per MWh captures 1 tonne of CO2 using 2.5 MWh of energy. What is the net carbon removal per tonne captured, and what does this imply about energy source selection?

CONSTRUCTED RESPONSE:

Using evidence from your model, argue whether direct air capture should receive more or less investment compared to renewable energy deployment for climate change mitigation. Address the energy requirement, cost comparison, and the moral hazard argument in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Humanity has emitted over 1.5 trillion tonnes of CO_2 since the Industrial Revolution, and we continue adding approximately 37 billion tonnes per year. Even if we stopped all emissions today, the CO_2 already in the atmosphere would keep warming the planet for decades. This reality has driven interest in technologies that can actively remove CO_2 from the atmosphere — essentially running the combustion reaction in reverse. Direct air capture is the most technologically mature approach, but it faces daunting challenges of thermodynamics, energy, cost, and scale.

How Direct Air Capture Works

DAC systems draw ambient air through chemical sorbents that selectively bind CO_2. In solid sorbent systems (like those used by Climeworks), air passes over porous materials coated with amine compounds that chemically bond with CO_2 at ambient temperature. Once saturated, the sorbent is heated to 80-120 degrees Celsius, releasing concentrated CO_2 for collection and storage. The sorbent cools and the cycle repeats. In liquid solvent systems (like Carbon Engineering's approach), air contacts a potassium hydroxide solution that absorbs CO_2 to form potassium carbonate. The carbonate is processed through a calcium caustic recovery loop that regenerates the solvent at 900 degrees Celsius and produces pure CO_2. Both approaches work but face the same fundamental challenge: CO_2 is only 420 ppm in air — a haystack-to-needle ratio that demands enormous energy.

The Thermodynamic Challenge

The minimum theoretical energy to separate CO_2 from air at 420 ppm is approximately 250 kWh per tonne, dictated by the second law of thermodynamics. Real systems require 1,500-2,500 kWh per tonne — 6-10 times the theoretical minimum — because of heat losses, air movement energy, material inefficiencies, and process engineering constraints. For comparison, capturing CO_2 from a coal plant's exhaust (at 120,000 ppm) requires only about 250-400 kWh per tonne total because the concentration is 300 times higher. This concentration difference is the fundamental reason DAC is far more expensive and energy-intensive than point-source capture — and it cannot be engineered away because it is a consequence of physics, not technology.

The Scale Problem

Current global DAC capacity is approximately 10,000-50,000 tonnes of CO_2 per year across all operational plants combined. To remove just 1% of annual emissions (370 million tonnes) would require scaling by a factor of 10,000 or more. The energy required would be equivalent to approximately 5-10% of current global electricity generation — energy that must come from zero-carbon sources or the system adds more CO_2 than it removes. The physical infrastructure would include thousands of large facilities, millions of tonnes of sorbent materials, and a completely new pipeline and injection industry for CO_2 storage. This is not technically impossible, but it represents an industrial buildout comparable to the entire global petroleum industry, constructed in decades rather than a century.

The Cost and Policy Landscape

Current DAC costs range from $400-$1,000 per tonne of CO_2, compared to the social cost of carbon estimated at $50-$200 per tonne and carbon offset prices of $5-$50 per tonne. For DAC to be economically viable at scale, costs must decrease by 5-10 times through improved sorbents, cheaper clean energy, and manufacturing scale effects. Government policies are beginning to support this: the US Inflation Reduction Act provides $180 per tonne tax credits for DAC with geological storage, and the Department of Energy has funded regional DAC hubs. Climeworks opened the world's largest DAC plant (Mammoth) in Iceland in 2024, capturing 36,000 tonnes per year using geothermal energy and storing CO_2 in basalt formations where it mineralizes within 2 years. These are encouraging steps but represent less than 0.0001% of the needed scale.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six key components of a direct air capture system: energy source carbon intensity, sorbent efficiency, thermal energy requirement, net carbon removal, capture cost, and deployment scale.

E **Establish Relationships**

Students map relationships showing that energy source carbon intensity is the determining variable for whether the system achieves net carbon removal, while sorbent efficiency and thermal requirements determine cost, and deployment scale reveals the enormous gap between current capacity and climate relevance.

V **Visualize & Model**

Students build a computational model showing the six-component DAC system with energy balance calculations and cost projections.

E **Evaluate Outcomes**

Students run renewable-powered, fossil-powered, and gigatonne-scale scenarios to discover the conditions under which DAC is effective, counterproductive, or feasible.

R **Revise & Extend**

Students add water consumption, sorbent lifetime, or storage permanence to model more realistic DAC deployment challenges and constraints.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"There is too much CO2 in the atmosphere. What if we could just build machines to suck it out? Turns out we can — but the numbers behind what it would take to actually matter are staggering."*

Watch For

Open with the simple premise: too much CO2, build machines to remove it. Then reveal the scale challenge: 420 ppm means only 0.042% of air is CO2. Ask: how would you extract something that dilute?

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling a technology that could be essential for climate stability — or could be a dangerous distraction from emission reductions. The data from your model will tell you which."*

Watch For

Have students read objectives. Pre-teach 'thermodynamic minimum energy' as the physics-imposed floor on extraction cost. Pre-teach 'gigatonne scale' as the target that matters for climate.

Slide 3: Big Question

3 min

Say: *"We CAN. Several companies are doing it right now. The question is not whether it works — it is whether it can ever work at a scale that matters for the climate, at a cost society can afford, without using so much energy that it makes the problem worse."*

Watch For

Quick-write: Students estimate how many tonnes of CO2 the world emits per year and how much a machine that captures one tonne would cost. Reveal the real numbers and watch reactions.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where the laws of thermodynamics set hard floors on energy requirements, and where the most critical variable is not the capture technology but the energy source powering it."*

Watch For

Preview LEVER steps. Emphasize that this model will quantify whether DAC is a climate solution or a climate delay tactic.

Slide 5: Activity 1: Components

8 min

Say: *"Six components. Two you control: energy source and deployment scale. Four that physics and economics determine. The most important insight: if your energy source emits CO2, you might be making the problem worse, not better."*

Watch For

Guide through components. Discuss why energy source and deployment scale are external choices. Sorbent efficiency, thermal requirements, net removal, and cost are consequences of physics and engineering.

Slide 6: Activity 2: Connections

8 min

Say: *"Energy source controls everything. With clean energy, DAC removes carbon. With dirty energy, DAC can add carbon. The same machine, the same chemistry, completely opposite outcomes depending on where the electricity comes from."*

Watch For

Help students map the critical energy source pathway. Calculate the break-even point: how clean does energy need to be for net removal? Highlight the cost-scale tension.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three scenarios, three very different stories. Let us find out whether DAC saves the climate, wastes money, or makes things worse."*

Watch For

Students predict net carbon removal before each scenario. The fossil-powered scenario is the critical insight — students realize the energy source can flip the system from solution to problem. The scale scenario reveals the infrastructure challenge.

Slide 8: Discoveries

5 min

Say: *"DAC is real, it works, and we probably need it. But it is not a magic eraser for emissions. It is expensive, energy-intensive, and cannot replace the urgent need to stop emitting CO2 in the first place."*

Watch For

Lead discussion on the moral hazard: does promising future carbon removal reduce today's urgency to cut emissions? Connect to real policy debates about carbon offsets and net-zero pledges.

Slide 9: STEM Challenge

12 min

Say: *"Your state wants net-zero by 2040 and is counting on DAC as one strategy. Design the deployment plan — but you have to show the numbers work."*

Watch For

Groups design regional DAC strategies with specific sites, energy sources, storage locations, and cost projections. Must address whether the investment would be better spent on emission reductions. Present with model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Energy Source Carbon Intensity, Deployment Scale
- **INTERNAL:** Sorbent Efficiency, Thermal Energy Requirement, Net Carbon Removal, Capture Cost

Relationships

- **Energy Source Carbon Intensity to Net Carbon Removal:** NEGATIVE (-) — Higher carbon intensity of the energy source means more CO2 is emitted to power the capture process, directly reducing and potentially eliminating or reversing the net carbon removal. At sufficiently high carbon intensity, the system adds more CO2 than it captures.
- **Sorbent Efficiency to Thermal Energy Requirement:** NEGATIVE (-) — More efficient sorbents capture more CO2 per cycle, meaning fewer cycles are needed to capture a given amount. Fewer cycles means less total thermal energy for sorbent regeneration. Higher efficiency sorbents also tend to release CO2 at lower temperatures, further reducing energy needs.
- **Deployment Scale to Capture Cost:** NEGATIVE (-) — Larger deployment scale enables manufacturing economies of scale, shared infrastructure, optimized supply chains, and learning-by-doing improvements that reduce the cost per tonne of CO2 captured — the same cost reduction pattern seen in solar panels and wind turbines over the past two decades.
- **Thermal Energy Requirement to Capture Cost:** POSITIVE (+) — Higher thermal energy requirements mean more energy must be purchased or generated per tonne of CO2 captured, directly increasing the operational cost. Energy is typically the largest single cost component in DAC operations.

Key Discoveries

- The diluteness of atmospheric CO2 (420 ppm) is the fundamental challenge — extracting something present at 0.042% concentration requires processing enormous volumes of air and consumes far more energy than capturing from concentrated industrial sources
- Energy source is the make-or-break variable — fossil-powered DAC can actually INCREASE atmospheric CO2 if the energy emissions exceed the capture amount, making renewable or nuclear energy essential for net-negative carbon removal
- Current costs of $400-$1,000 per tonne must decrease by an order of magnitude for DAC to compete with other climate strategies — for reference, the social cost of carbon is estimated at $50-$200 per tonne
- Scaling DAC to remove even 10% of annual emissions would require energy equivalent to a significant fraction of current global electricity generation — raising questions about whether that energy would be better used replacing fossil fuels directly

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Direct air capture can solve climate change so we do not need to reduce emissions"	DAC is orders of magnitude too small, too expensive, and too energy-intensive to substitute for emission reductions. Current global DAC capacity removes less than 0.001% of annual emissions. Even optimistic projections for 2050 target 5-10 gigatonnes per year against 37+ gigatonnes of ongoing emissions. Furthermore, the energy required for gigatonne-scale DAC would consume a significant fraction of global clean energy production — energy better used to directly replace fossil fuels. The model clearly shows that DAC is a complement to emission reductions, not a substitute.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Planting trees is just as good as direct air capture"	Trees are excellent carbon sinks but face limitations that DAC does not: they require land (competing with agriculture), take decades to reach full carbon absorption, are vulnerable to fire, drought, and disease that release stored carbon, and have a finite saturation point. A mature forest eventually reaches carbon equilibrium where absorption equals decomposition. DAC can operate on non-arable land, captures CO2 at a constant rate regardless of season, and stores carbon as permanent minerals rather than temporary biomass. However, trees cost $5-$50 per tonne versus $400-$1,000 for DAC — so where forests are viable, they are far more cost-effective.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Carbon capture technology is just greenwashing by fossil fuel companies"	This is partially correct and partially wrong. Some fossil fuel companies do promote carbon capture as a way to justify continued fossil fuel production — this is a legitimate criticism. However, the technology itself is scientifically valid and may be genuinely necessary for addressing historical emissions and hard-to-decarbonize sectors. The critical distinction is between using DAC to enable continued fossil fuel extraction (greenwashing) versus using DAC to clean up residual emissions in a world that is rapidly decarbonizing (legitimate climate strategy). The model helps students distinguish between these uses by showing that DAC only achieves net removal when powered by clean energy.